

Module 1: Sets and Set Theory

- ✓ **Course structure and navigation**
Reading • 15 min
- ✓ **How to learn effectively on this course**
Reading • 15 min
- **Introduction to the specialisation and this course**
Video • 4 min
- **Course Syllabus**
Reading • 10 min

- The definition of a set
Video • 8 min
- The definition of a set
Reading • 10 min
- The definition of a set
Practice Assignment • 30 min
- The listing method and rule of inclusion
Video • 8 min
- The listing method and rule of inclusion
Practice Assignment • 30 min
- The powerset of a set
Video • 10 min
- The powerset of a set
Practice Assignment • 15 min
- Set operations
Video • 10 min
- Set operations
Practice Assignment • 30 min

- The representation of a set using Venn diagrams
Video • 4 min
- The representation of a set using Venn diagrams
Practice Assignment • 30 min
- Mastering Set Operations and Venn Diagram Reasoning
Dialogue • 30 min
- De Morgan's laws
Video • 8 min
- De Morgan's laws
Practice Assignment • 30 min
- Laws of sets: Commutative, associative and distributives
Video • 11 min
- Laws of sets: Commutative, associative and distributives
Reading • 15 min
- Laws of sets: Commutative, associative and distributive
Practice Assignment • 30 min
- Partition of a set
Video • 3 min
- Partition of a set
Practice Assignment • 30 min

- Module 1 summary
Reading • 15 min
- Check your understanding: End of module 1
Graded Assignment • 20 min

Module 3: Functions and Their Properties

Lesson 4.1. Understanding the concept of relations

- **Definition of a relation: relation versus function**
Video • 8 min
- **Definition of a relation: relation versus function**
Practice Assignment • 20 min
- **Matrix and graph representations of a relation**
Video • 10 min
- **Matrix and graph representations of a relation**
Practice Assignment • 20 min
- **The properties of a relation: reflexive, symmetric and anti-symmetric**
Video • 14 min
- **The properties of a relation: reflexive, symmetric and anti-symmetric**
Practice Assignment • 20 min
- **Relation properties: transitivity**
Video • 5 min
- **Relation properties: transitivity**
Practice Assignment • 20 min

- **Equivalence relations and equivalence classes**
Video • 6 min
- **Equivalence relations and equivalence classes**
Practice Assignment • 25 min
- **Partial and total order**
Video • 6 min
- **Partial and total order**
Practice Assignment • 30 min
- **Building and analysing different types of mathematical relations**
Dialogue • 15 min

- Topic summary
Reading • 15 min
- Check your understanding: End of module 4
Graded Assignment • 20 min

Mathematical Foundations for Computing:
Course Summary
Reading • 10 min

The Archive Room: A Mathematical Escape
Mystery
Role Play • 15 min

Course summary
Video • 1 min

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Staying motivated

To make your learning habit stick, set a fixed study schedule and pair it with an existing routine, like watching a video with your morning coffee or reviewing notes before bed. Linking a new habit to something you already do makes it easier to remember and follow through – this is called **habit stacking**. In addition, use the platform's tools to track your progress, for example, click 'Mark as complete' for each activity so you can easily see how far you've come. Staying accountable also helps. Connect with others in discussion forums or ask a friend or a family member to check in on your progress. And remember, motivation grows when learning feels meaningful – try connecting what you're learning to real-world topics that matter to you.

Learning to learn

In fact, learners who test themselves, whether with closed or open-ended questions or by explaining ideas in their own words, retain far more than those who simply review. This approach is especially powerful because of something called the **forgetting curve**, which shows that we begin to forget information almost immediately after learning it. When you test yourself, you interrupt that forgetting curve and help your brain hold on to what matters. Coming back to the material again later strengthens learning even more, this means revisiting what you've learned after a break, instead of trying to memorise it all at once.

If you're interested in exploring the concepts behind these strategies, such as, **retrieval practice**, **spaced practice**, **elaboration**, **metacognition**, **interleaving** and **concrete examples**, you're more than welcome to. However, that's not the intention of this course. You don't need to memorise these terms; what's most important is understanding that **how** you study is just as important as **what** you study.

We commend you to start your own study journal, in a format of your choosing, which could be traditional pen and paper or an electronic version. Your journal will enable you to collate information, monitor progress and organise your notes. You can use it later for reference when completing graded assignments.

This is for your reference only and will not be assessed.

The format that you are most comfortable with!

You may prefer to use pen and paper, but it is worth bearing in mind that electronic documents are easily searchable and can store links.

Online documents are preferable to static documents as you will be able to access them from any computer – handy if you study in more than one place! Options include a live Word doc, Google doc, OneNote, a personal blog, Evernote or any other document of your choice.

Journal entries could include the following:

- Links to websites and articles
- Your own thoughts and reflections
- Your responses to directed activities
- Notes on readings and videos.

To give you an idea of how you could set yours out, here is [a screencast example of a study journal](#) .

Start your own study journal using one of the tools suggested.

Now post an entry in your study journal answering the following questions:

- How will you record reflective practice in your journal?
- Which journal tool will you use?

Let's get started on the course.

Go to next item ✓ Completed